

Arco Instrument Design Specification

Version: 0.1

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Repo: <https://github.com/camille-004/arco>

What is Arco?

Arco is a desktop electronic instrument that you play by “bowing”, with your finger sliding along a touch strip. The left strip selects the note. The right strip is the bow. How fast you slide determines how much energy enters the sound. Stop moving and the sound fades, just like lifting a bow from a string.

The name comes from the Italian musical term *arco* (“play with the bow”), used in scores to tell string players to return to bowing after plucking.

How It Sounds

Arco uses a physically modeled string, a mathematical simulation of a vibrating resonant body. The sound is in the territory of a bowed instrument: sustained, expressive, with a natural attack and decay. It is not a synthesizer pad or a sampler. The timbre responds continuously to how you play, instead of the knobs you turn.

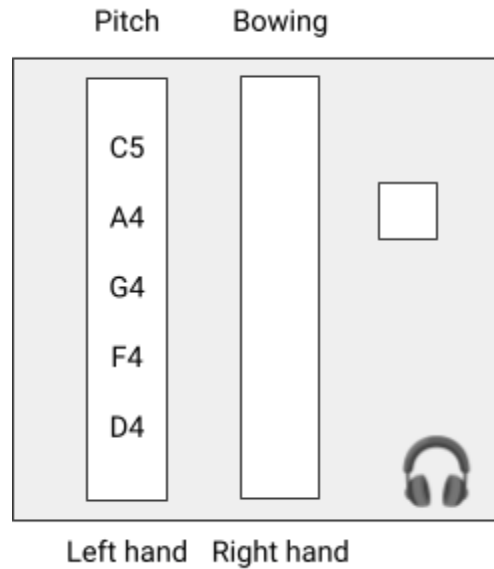
Two presets shape the character of the resonance:

Preset	Character
Staccato	Shorter decay, driver, more articulate. Notes speak quickly and fade.
Sustained	Long ring, wetter reverb, legato.

A single button on the top face cycles between them.

How You Play It

Arco sits flat on a desk. Two touch strips run across the top face side by side.



Left hand: pitch. The strip is divided into five zones, one per note of the D minor pentatonic scale. Press a zone to select that note. Moving between zones triggers a fresh attack at the new pitch.

Zone	Note	Frequency (Hz)
1 (bottom)	D4	293.66
2	F4	349.23
3	G4	392.00
4	A4	440.00
5 (top)	C5	523.25

The D minor pentatonic scale was chosen for its expressive, melancholic character. Any combination of its notes sounds coherent, which makes it forgiving to learn on while technique develops.

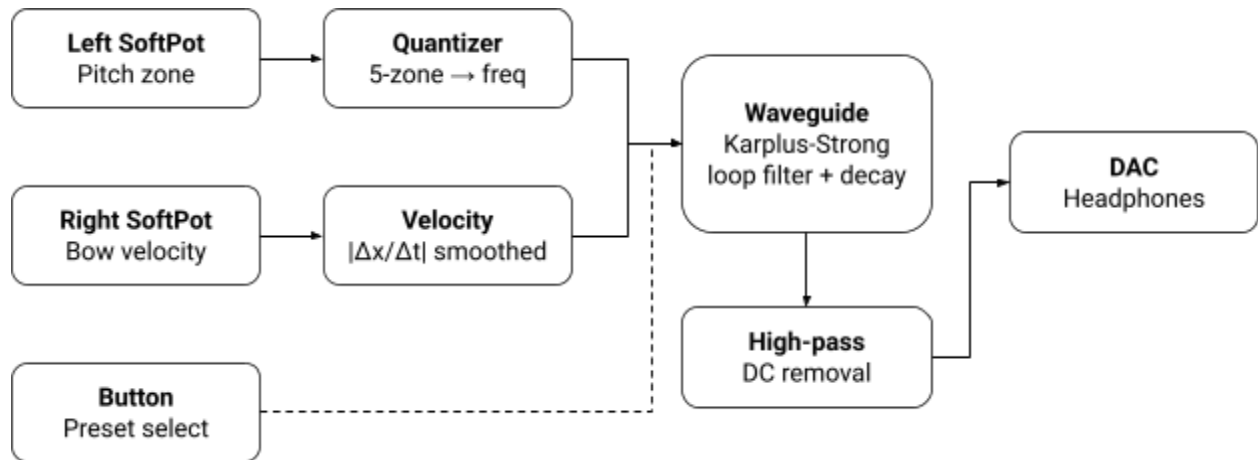
Right hand: bowing. Slide your finger along the strip. The speed of your movement, rather than where your finger is, determines how much energy enters the sound. Slow, controlled sliding produces a gentle sustained tone. Fast sliding produces a louder, richer, more harmonically complex sound. Holding still lets the sound decay naturally.

This distinction matters because position-based control is a parameter knob, and velocity-based control is physical energy input, which is the same relationship a bow has with a string.

The Sound Engine

For those curious about what's happening under the hood.

Arco uses [Karplus-Strong digital waveguide synthesis](#)—one of the foundational techniques in physical modeling, developed at Stanford in 1983. The core idea is a circular buffer of audio samples with a feedback loop. The length of the buffer sets the pitch (shorter = higher, like a shorter string). Each time a sample travels around the loop, it passes through a filter that removes a little energy, modelling the natural damping of a real string. The result rings at the resonant frequency of the buffer.



The pitch period in samples is:

$$N = \frac{f_s}{f_0}$$

Where $f_s = 44100\text{Hz}$ and f_0 is the note frequency. The loop filter is a simple weighted average:

$$y[n] = (1 - \beta) \cdot \frac{x[n] + x[n+1]}{2} + \beta \cdot x[n]$$

β is the brightness parameter. Lower values produce a darker, more fundamental-heavy tone; higher values let harmonics through. While bowing, a small continuous noise signal is injected into the loop to sustain the oscillation, scaled by bow velocity v :

$$x[n] \leftarrow \delta \cdot x[n] + v \cdot \varepsilon[n], \varepsilon[n] \sim U(-0.06, 0.06)$$

Where δ is the per-sample decay coefficient. Reverb is a single [feedback comb filter](#):

$$c[n] = s[n] + \alpha \cdot c[n - L]$$

With $\alpha = 0.82$, $L \approx 43ms$.

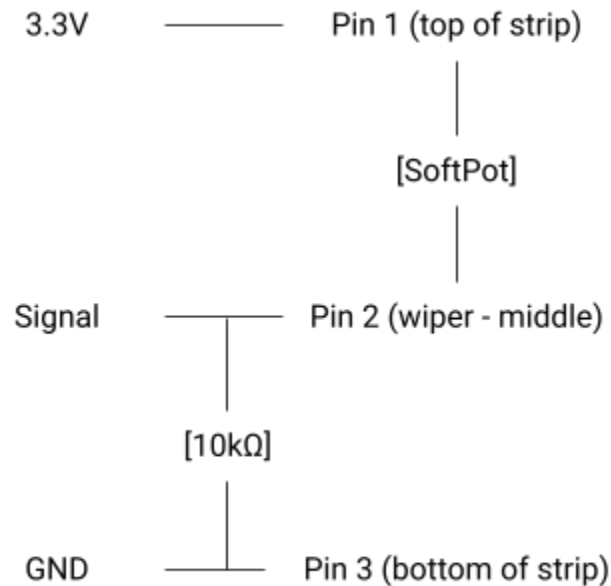
Reference: Julius O. Smith, [Physical Audio Signal Processing](#), CCRMA Stanford.

Parts

Component	Where to buy	~Price (\$)
Teensy 4.0 (with headers)	SparkFun	24
Audio Shield Rev D	SparkFun	17
SoftPot 50mm x2	SparkFun	18
Stackable header kit	SparkFun	2
10kΩ resistors x2	Amazon	2
Tactile pushbutton	Amazon	2
ABS project enclosure	Amazon	8
Total		73

The Teensy 4.0 is a small, powerful microcontroller running at 600 MHz, which is fast enough to do real-time audio synthesis. The Audio Shield adds a high-quality stereo headphone output. The SoftPots are thin flexible touch strips that output a voltage proportional to where you press them.

Wiring



The Audio Shield stacks directly onto the Teensy. This is the one step that requires soldering. Female header pins are soldered to the shield once, after which the Teensy plugs in and out freely. Everything else connects with breadboard jumper wires.

Wiring reference: [SparkFun SoftPot Hookup Guide](#)

Build Phases

Prototype

Sound validated in Python. Mouse simulates bowing, keyboard simulates pitch. Both presets confirmed working.

Hardware

Wire SoftPots and button to Teensy on a breadboard. Port the waveguide synthesis to the Teensy Audio Library. Confirm the sound matches the prototype.

Tuning

Tune the bow velocity response and pitch zone boundaries to feel right on real hardware. Adjust preset parameters as needed.

Enclosure

Mount everything in the ABS box. Cut openings for the SoftPots, button, and headphone jack.

References.

- Smith, J. O. (2010). *Physical Audio Signal Processing*.
<https://ccrma.stanford.edu/~jos/pasp/>
- Karplus, K. & Strong, A. (1983). Digital synthesis of plucked-string and drum timbres. *Computer Music Journal*, 7(2). <https://doi.org/10.2307/3679991>
- SparkFun SoftPot Hookup Guide.
<https://learn.sparkfun.com/tutorials/softpot-hookup-guide/all>
- PJRC Teensy Audio Library. https://www.pjrc.com/teensy/td_libs_Audio.html